Cab Switchbox Manual

> ESP32 network switch input setup, wiring, and troubleshooting

Last updated: June 19, 2026

Hardware shown in this manual: **SwitchBox5-Terminals-PCB-01 (Rev 01)**.

Overview

The ESP32 Cab Switchbox is a network switch input module for the rate controller system. It reads up to 16 cab switch inputs, converts them into the switchbox packet used by the rate controller, and sends that packet over UDP by Ethernet and WiFi.

The switchbox is intended for cab controls such as:

- Master on / master off
- Rate up / rate down
- Manual rate override
- Manual section override
- Rate 2
- Work switch
- Section switches 1 through 16
- Board-mounted alarm buzzer output

The current firmware is in:

```
ESP32_Cab_Switchbox/ESP32_Cab_Switchbox.ino
```

Default Network Settings

The switchbox starts its own WiFi access point and can also use Ethernet.

Default settings:

Item	Default
Access point name	`CabSwitchbox-XXXX`
Access point password	blank / open network
WiFi setup IP	`192.168.0.71`
Ethernet IP	`192.168.0.70`
Gateway	`192.168.0.1`
Subnet	`255.255.255.0`
Web page port	`80`
mDNS name	`cabswitchbox.local`
UDP send port	`29999`
Local UDP port	`28888`

Additional Parts Needed

The SwitchBox5 Rev 01 terminal PCB needs these additional parts installed before use:

Part	Notes
2 amp mini fuse	Installs in the fuse holder. Protects the input/switch-feed circuit.
ESP32-DEVKitC-VE	Main controller board. Install in the ESP32 header sockets.
W5500 Ethernet Module	Ethernet network module. Install in the W5500 module header.
Hammond `E1555K2GY` enclosure	Polycarbonate NEMA 4X enclosure, 6.3 x 3.5 x 3.5 inches. The SwitchBox5 PCB fits this enclosure.

Enclosure

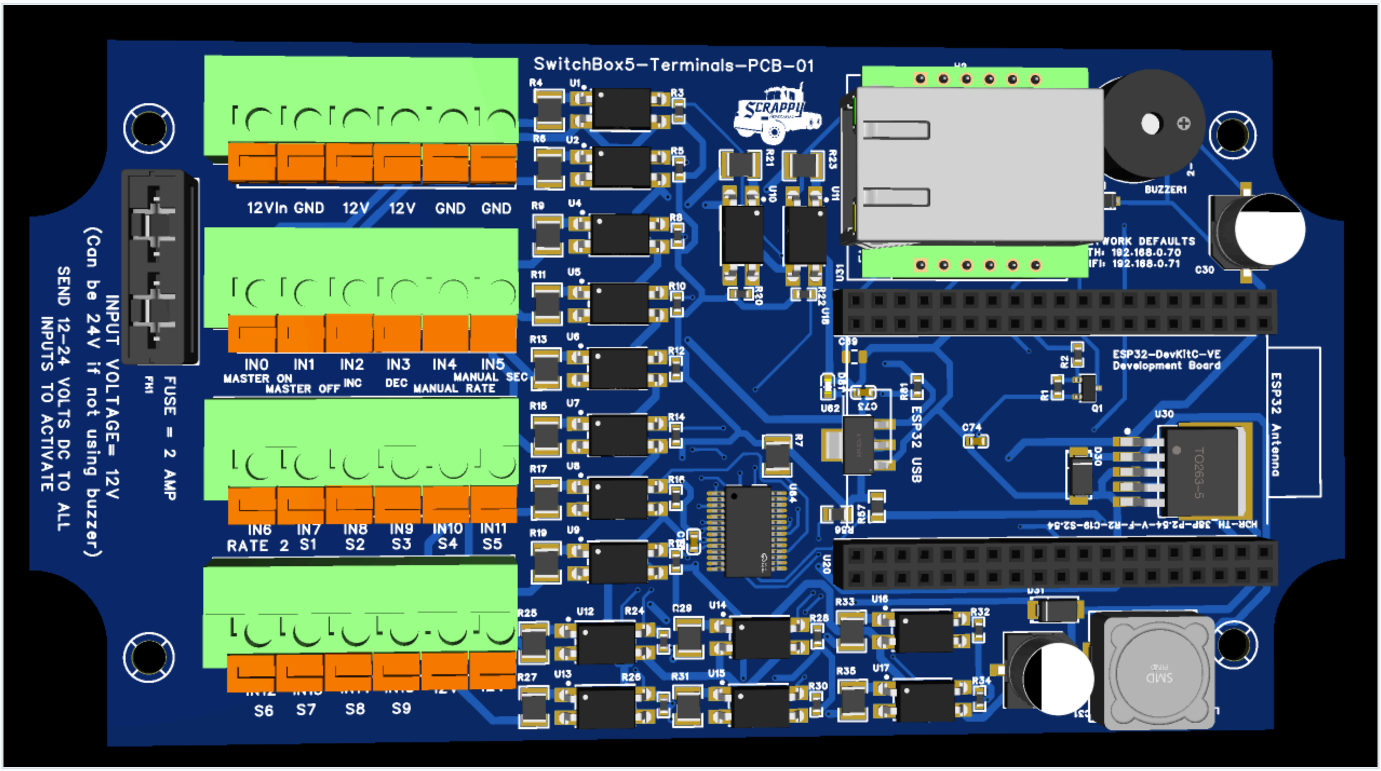
The SwitchBox5 PCB fits the **Hammond Manufacturing E1555K2GY**

polycarbonate enclosure:

Item	Specification
Manufacturer	Hammond Manufacturing
Part number	`E1555K2GY`
Outside size	6.3 x 3.5 x 3.5 inches
Protection rating	NEMA 4X
Material	Polycarbonate

Plan cable glands, Ethernet access, switch wiring, and board mounting before drilling the enclosure. Keep the lid sealing surface clear of wiring and hardware so the NEMA 4X seal is maintained.

Board Power And Switch Feed



SwitchBox5-Terminals-PCB-01 (Rev 01). Connect board power to 12Vin and GND. Use the fused 12V terminals through switches to power the IN0-IN15 inputs.

Power And Switch Feed Terminals

Power the board from the input power terminals:

The upper six-position terminal block on Rev 01 is ordered left to right:

Position	Terminal	Use
1	`12Vin`	Connect to supply +12-24 VDC.
2	`GND`	Connect to supply negative / ground.
3	`12V`	Fused +12-24 VDC output for switch feeds.
4	`12V`	Fused +12-24 VDC output for switch feeds.
5	`GND`	Ground return terminal.
6	`GND`	Ground return terminal.

The `12V` terminals are fused outputs from the board input power. Use these fused `12V` terminals to feed the operator switches, then return the switched voltage to the desired `INx` input.

Typical wiring:

```
Supply +12-24 VDC -------- 12Vin

Supply negative ---------- GND



Fused 12V terminal ------- switch ------- INx
```

The Rev 01 input power fuse is 2 amps. The fused `12V` terminals are intended for feeding switch inputs, not for powering heavy loads.

The PCB electronics and switch inputs can operate from **12-24 VDC**. However,

the buzzer is a **12 V board-mounted component connected directly to input

power**. The buzzer will only operate correctly when the board is powered from

12 VDC. Do not expect buzzer operation with a 24 V supply, and do not apply

24 V to the installed 12 V buzzer.

Rev 01 Terminal Layout

The four six-position terminal blocks are ordered as follows:

Block	Left-to-right terminal order
Power	`12Vin`, `GND`, `12V`, `12V`, `GND`, `GND`
Inputs 0-5	`IN0`, `IN1`, `IN2`, `IN3`, `IN4`, `IN5`
Inputs 6-11	`IN6`, `IN7`, `IN8`, `IN9`, `IN10`, `IN11`
Inputs 12-15 / switch feed	`IN12`, `IN13`, `IN14`, `IN15`, `12V`, `12V`

The Rev 01 silkscreen also identifies the default functions below the input

terminals: master on/off, rate increase/decrease, manual rate, manual section,

Rate 2, and sections S1 through S9.

Accessing The Setup Page

Using The Switchbox WiFi Access Point

1. Power the Cab Switchbox.

2. On a phone, tablet, or laptop, connect to the WiFi network named `CabSwitchbox`.
3. Open a browser and go to:

```
http://192.168.0.71/
```

If mDNS is working on the device you are using, this may also work:

```
http://cabswitchbox.local/
```

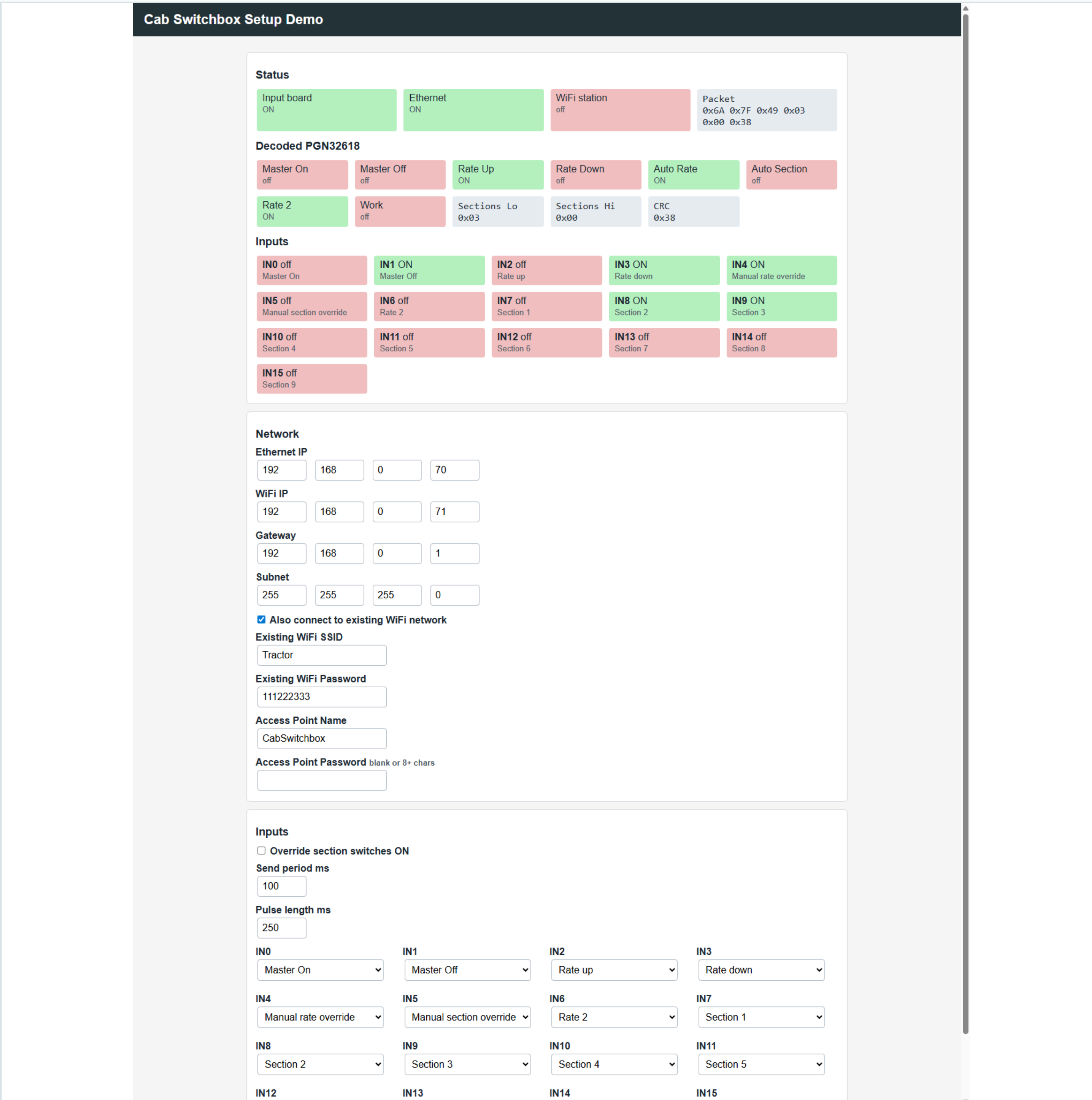
Using Ethernet

1. Connect the switchbox Ethernet port to the same network as the rate controller / setup computer.
2. Make sure the computer is on the same subnet as the switchbox.
3. Open:

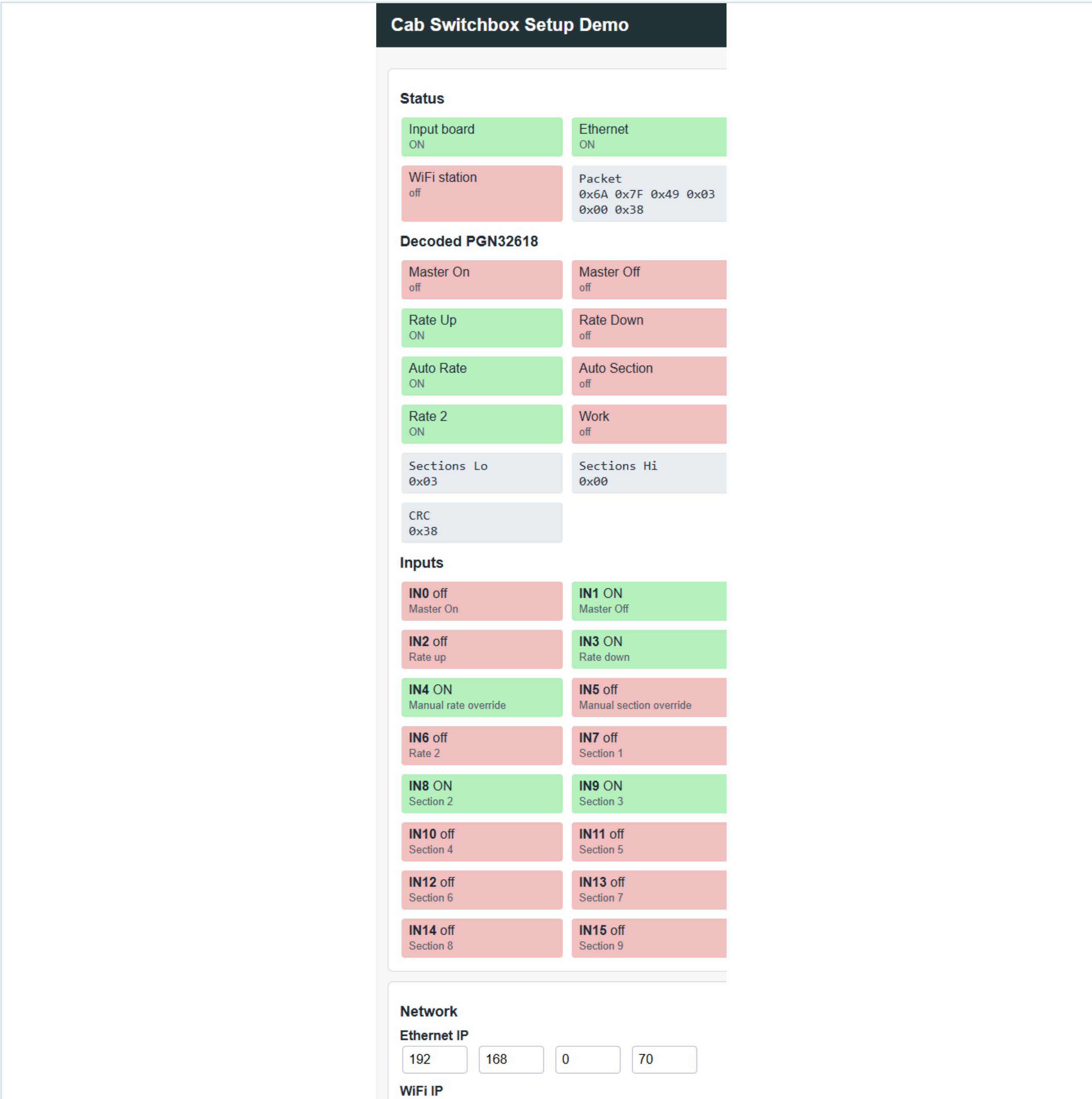
```
http://192.168.0.70/
```

If needed, first connect by WiFi and change the Ethernet IP address to match the network you are using.

Setup Page Screenshots



Desktop setup page with live status, decoded packet, network settings, and input configuration.



Mobile layout used when configuring the switchbox from a phone or tablet in the cab.

Setup Page Sections

The setup page has a live status area and a configuration form. After changing settings, click **Save and restart**. The switchbox saves the settings to EEPROM and restarts.

Status

The status area shows:

Status item	Meaning
Input board	The switch input board is detected and ready.
Ethernet	Ethernet hardware is detected and the link is up.
WiFi station	The switchbox is connected to an existing WiFi network.
Buzzer	The GPIO27 buzzer output is currently on.
Alarm packet	A recent alarm packet has been received from Rate Controller.
Alarm active	Rate Controller is requesting the cab buzzer.
Packet	The six-byte packet being sent to the rate controller.
Decoded PGN32618	Human-readable state of the outgoing switchbox packet.
Inputs	Live state and assigned action for each input `IN0` through `IN15`.

Network Settings

Setting	Description
Ethernet IP	Static IP used by the Ethernet interface.
WiFi IP	IP used by the switchbox access point.
Gateway	Gateway used for Ethernet and station WiFi.
Subnet	Subnet mask.
Also connect to existing WiFi network	Enables WiFi station mode in addition to the switchbox access point.
Existing WiFi SSID	Name of the WiFi network the switchbox should join.
Existing WiFi Password	Password for the existing WiFi network.
Access Point Name	Name of the WiFi network created by the switchbox.
Access Point Password	Password for the switchbox access point. Leave blank for open WiFi, or use 8 or more characters.

Input Settings

Setting	Default	Description
Override section switches ON	Off	Forces mapped section switch inputs ON without jumpers. Use this when section switches are not installed.
Send period ms	`100`	How often the switchbox packet is sent. Valid range is 50 to 1000 ms.
Pulse length ms	`250`	Length of master and rate pulses. Valid range is 50 to 1000 ms.
IN0-IN15 action	See table below	Assigns what each physical input does.

Alarm Buzzer Settings

Setting	Default	Description
Buzzer level	`Reduced`	Select `Off`, `Reduced`, or `Full` for the board buzzer.
Test buzzer	n/a	Sounds the buzzer for two seconds without changing saved settings.

Default Input Map

Factory/default input assignments:

Input	Default action	Typical wiring
IN0	Master on	Momentary switch feeding +12-24 VDC to IN0
IN1	Master off	Momentary switch feeding +12-24 VDC to IN1
IN2	Rate up	Momentary switch feeding +12-24 VDC to IN2
IN3	Rate down	Momentary switch feeding +12-24 VDC to IN3
IN4	Manual rate override	Toggle switch feeding +12-24 VDC to IN4
IN5	Manual section override	Toggle switch feeding +12-24 VDC to IN5
IN6	Rate 2	Toggle switch feeding +12-24 VDC to IN6
IN7	Section 1	Toggle switch feeding +12-24 VDC to IN7
IN8	Section 2	Toggle switch feeding +12-24 VDC to IN8
IN9	Section 3	Toggle switch feeding +12-24 VDC to IN9
IN10	Section 4	Toggle switch feeding +12-24 VDC to IN10
IN11	Section 5	Toggle switch feeding +12-24 VDC to IN11
IN12	Section 6	Toggle switch feeding +12-24 VDC to IN12
IN13	Section 7	Toggle switch feeding +12-24 VDC to IN13
IN14	Section 8	Toggle switch feeding +12-24 VDC to IN14
IN15	Section 9	Toggle switch feeding +12-24 VDC to IN15

Any input can be reassigned from the web page.

Available actions:

Action	Behavior
Disabled	Input is ignored.
Master on	Momentary master on pulse.
Master off	Momentary master off pulse.
Rate up	Rate up command.
Rate down	Rate down command.
Work switch	Implement up/down permissive. ON allows application; OFF forces master/sections off when Work Switch is enabled in the Rate Controller app.
Manual section override	When active, requests manual section override.
Manual rate override	When active, requests manual rate override.
Rate 2	Sends Rate 2 while active.
Master	Maintained master switch. ON sends master on; OFF sends master off.
Section 1-16	Sends the matching section switch bit while active.

Wiring Inputs

The switchbox5 board inputs are made for 12-24 VDC switch signals. Each switch input turns ON when +12-24 VDC is applied to that `INx` terminal.

Input behavior:

- An open / unpowered input reads OFF.
- Applying +12-24 VDC to an `INx` terminal reads ON.
- The 12-24 VDC supply negative must be connected to the switchbox input `GND`.

Typical switchbox5 field wiring:

```
12Vin ---------------- supply +12-24 VDC

GND ------------------ supply negative

12V fused output ------ switch ------ INx
```

For a momentary button, use a normally-open button between a fused `12V` terminal and the assigned `INx` terminal.

For a toggle switch, use a maintained switch between a fused `12V` terminal and the assigned `INx` terminal.

Use the `IN0` through `IN15` input terminals on the switchbox board. Do not connect field wiring to internal board components.

Wiring The Master And Rate Controls

A simple installation without section switches can use only:

Function	Default input
Master on	IN0
Master off	IN1
Rate up	IN2
Rate down	IN3
Manual rate override	IN4
Manual section override	IN5
Rate 2	IN6

Wire each installed switch so it feeds fused +12-24 VDC from a `12V` terminal to the assigned `INx` terminal when the switch is ON or pressed.

If section switches are not installed, enable **Override section switches ON**. This makes all inputs assigned to `Section 1` through `Section 16` behave as ON in the outgoing packet, so the system does not require jumper wires on the unused section switch inputs.

Wiring Section Switches

For normal section switch operation:

1. Leave **Override section switches ON** unchecked.
2. Wire each section toggle so it feeds fused +12-24 VDC from a `12V` terminal to its assigned `INx` terminal when ON.

Default section switch wiring:

Section	Default input
Section 1	IN7
Section 2	IN8
Section 3	IN9
Section 4	IN10
Section 5	IN11
Section 6	IN12
Section 7	IN13
Section 8	IN14
Section 9	IN15

Sections 10 through 16 can be used by reassigning inputs from the setup page. The board has 16 total inputs, so using more section switches may require disabling or moving other functions.

Override Section Switches ON

Use **Override section switches ON** when the operator does not have physical section switches installed.

When enabled:

- Inputs assigned to `Section 1` through `Section 16` are forced ON in the switchbox packet.
- Master, rate, and override inputs still use the real physical switch state.
- No section input jumpers are required.

When disabled:

- Section inputs behave normally.
- An open / unpowered section input is OFF.

Work Switch

The `Work switch` input is normally used for an implement lift switch, height switch, whisker switch, pressure switch, or other signal that tells the rate controller whether the implement is in work.

Typical use:

```
Implement down / in work = +12-24 VDC to Work switch input = ON

Implement up / out of work = input unpowered = OFF
```

When Work Switch is enabled in the Rate Controller app, it acts as a master permissive:

- Work switch ON allows the rate controller to apply when the master and section conditions are also satisfied.
- Work switch OFF forces master/sections off.
- It does not replace Auto Rate or Auto Section. Those are separate switchbox functions.

In the Rate Controller app, Work Switch must be enabled in the switch settings. If it is not enabled there, the app ignores the work switch signal and behaves as if work is always allowed.

Alarm Buzzer Output

The active 12 V alarm buzzer is installed directly on the SwitchBox5 PCB. It is not an additional external part. The buzzer is driven from ESP32 Dev Board pin `GPIO27` through the onboard low-side MOSFET.

The buzzer receives board input voltage directly. Therefore:

- With a 12 VDC board supply, the onboard buzzer operates normally.
- The remainder of the switchbox can operate from 12-24 VDC.
- With a 24 VDC board supply, the installed 12 V buzzer must not be operated.

GPIO27 does not need an external pull-up. The 100k pulldown keeps the buzzer off while the ESP32 is booting or reset.

Packet Behavior

The switchbox sends a six-byte packet to UDP port `29999`.

Packet layout:

Byte	Meaning
0	Header byte `106`
1	Header byte `127`
2	Status bits
3	Section bits 1-8
4	Section bits 9-16
5	Checksum

Status byte bits:

Bit	Meaning
0	Rate 2
1	Master on
2	Master off
3	Rate up
4	Rate down
5	Auto section
6	Auto rate
7	Work switch

Section byte behavior:

- Byte 3 bit 0 = Section 1
- Byte 3 bit 7 = Section 8
- Byte 4 bit 0 = Section 9
- Byte 4 bit 7 = Section 16

The checksum is the sum of bytes 0 through 4, stored in one byte.

Recommended First Setup

1. Power the switchbox.
2. Connect to WiFi network `CabSwitchbox`.
3. Open `http://192.168.0.71/`.
4. Confirm the status page shows `Input board` ON.
5. If using Ethernet, set the Ethernet IP to match the rate controller network.
6. If section switches are not installed, check **Override section switches ON**.
7. Confirm each installed switch changes state in the Inputs status area.
8. Click **Save and restart**.
9. After restart, verify the rate controller responds to master and rate commands.

Example Setups

Master And Rate Only, No Section Switches

Use this when only master and rate controls are installed.

Recommended settings:

Setting	Value
Override section switches ON	Checked
IN0	Master on
IN1	Master off
IN2	Rate up
IN3	Rate down
IN4-IN6	As needed
IN7-IN15	Section 1-9, or Disabled

Full Section Switchbox

Use this when physical section switches are installed.

Recommended settings:

Setting	Value
Override section switches ON	Unchecked
IN0-IN6	Master/rate controls
IN7-IN15	Section 1-9 by default

Maintained Master Switch

If using a maintained master switch instead of separate master on/off buttons:

1. Set one input action to `Master`.
2. Wire the maintained switch so it feeds fused +12-24 VDC from a `12V` terminal to that input when ON.
3. Set unused `Master on` and `Master off` inputs to `Disabled`, or reassign them.

When the maintained master input turns ON, the switchbox sends master on. When it turns OFF, the switchbox sends master off.

Troubleshooting

Cannot Open The Web Page

- Make sure the switchbox is powered.
- If using WiFi, connect to the `CabSwitchbox` access point first.
- Try `http://192.168.0.71/` for WiFi access.
- Try `http://192.168.0.70/` for Ethernet access.
- If the IP address was changed, connect by the other interface or reflash/reset configuration if needed.

Input Board Shows OFF

The switchbox is not detecting the input board.

Check:

- Input board power and ground
- I2C SDA/SCL wiring
- Soldering around the input board

If Input board is OFF, switch input states will not be reliable.

Ethernet Shows OFF

Check:

- Ethernet cable
- Link lights on the W5500 module or connector
- Network switch/router
- W5500 SPI wiring and chip select

The switchbox can still be configured by WiFi if Ethernet is unavailable.

Inputs Do Not Turn ON

For the switchbox5 board, each input needs +12-24 VDC on the `INx` terminal to turn ON.

Check:

- The switch feed has +12-24 VDC.
- The switch output is connected to the correct `INx` terminal.
- The 12-24 VDC supply negative is connected to switchbox input `GND`.
- The input is assigned to the expected action in the setup page.

Missing Section Switches Stop Operation

Enable **Override section switches ON**. This forces section switch bits ON without changing master or rate input behavior.

Rate Up Or Rate Down Sticks ON

Check whether the input is assigned to `Rate up` or `Rate down` and whether the physical switch is stuck active. The status page shows live input states.

Check whether the physical switch is stuck on, or whether +12-24 VDC is being backfed into that input.

WiFi Station Does Not Connect

Check:

- SSID spelling
- Password spelling
- WiFi signal strength
- Whether the WiFi network uses a compatible 2.4 GHz mode

The switchbox access point remains available even when station WiFi is enabled.

Notes For Firmware Updates

Configuration is stored in EEPROM. The current firmware uses config version 3.

Defaults are restored when:

- EEPROM does not contain a valid switchbox config.
- The stored config version is not recognized.

The firmware includes migration for older switchbox config versions 1 and 2.